Meet D-Rex, a Giant Fungus Whose Family Began 65 Million Years Ago

From: ScienceDaily Strange & Offbeat Mined: 2026-10-27 5 min read

Scientists exploring forests in Yunnan, China, have discovered a giant fungus with a dinosaur-sized nickname: D-rex. Its upright reproductive structures can reach 50 cm (20 inches) tall and weigh as much as 2.2 kg (4.9 pounds), making them the largest known fruiting bodies among fungi in the phylum Ascomycota.

The species is called Dianjunus rex, a name inspired by its impressive size and the ancient Dian kingdom. Researchers said it is different enough from known fungi to need a new **genus** and an entirely new family, Dianjunaceae, within the order Xylariales.

The team found the fungus while surveying temperate coniferous forests near Damaidi and Yipinglang, and tropical rainforest near Nanbang. They spotted unusually large, whitish structures, known as **stromata**, growing from the woody roots of Chinese cork oak trees.

Ascomycetes make up the largest major group of fungi, with more than 64,000 recognized species. Unlike familiar mushrooms, they do not produce true mushroom structures. Instead, they make spores inside tiny sac-like structures found inside or on their fruiting bodies.

Using a method called a **molecular clock**, scientists compared the fungus's DNA with genomic information from hundreds of other fungi. Their results suggest the Dianjunaceae family first appeared approximately 65 million years ago, roughly one million years after the mass extinction that eliminated the non-avian dinosaurs.

That timing raises a possibility, said Dr Indunil C Senanayake, of the Kunming Institute of Botany: after the extinction, forests changed dramatically, and large amounts of dead plant material suddenly became available. For wood-associated **decomposing** fungi, he said, such disruption may have created new ecological opportunities.

Word Treasure

stromata -- The whitish, upright structures of this fungus that grow from tree roots.

Ascomycetes -- The largest major group of fungi, which make spores inside tiny sacs.

molecular clock -- A method that compares DNA to work out when a group of living things first appeared.

genus -- A group of closely related species, one step above a single species.

decomposing -- Breaking down dead plant or animal material, as fungi do.

Background you need

Fungi are living things that are neither plants nor animals, and many of them feed by breaking down dead wood and other plant material.

Unlike familiar mushrooms, ascomycetes make their spores inside tiny sac-like structures on or inside their fruiting bodies.

Scientists group living things into species, genera and families, and a fungus different enough from all known fungi may need a brand-new genus and family of its own.

By comparing DNA with a molecular clock, researchers can estimate when a family first appeared, even from a very long time ago.

Break it down

LEAD: Scientists in Yunnan, China, discovered a giant fungus nicknamed D-rex.

KEY EVENT: The fungus, Dianjunus rex, is different enough to need a new genus and an entirely new family, Dianjunaceae.

+- WHO: Researchers surveying forests, including Dr Indunil C Senanayake of the Kunming Institute of Botany

+- WHERE: Temperate coniferous forests near Damaidi and Yipinglang, and tropical rainforest near Nanbang

+- WHAT: Upright reproductive structures up to 50 cm tall and 2.2 kg, the largest known fruiting bodies among Ascomycota

EVIDENCE: A molecular clock comparison of its DNA with hundreds of other fungi suggests the family first appeared approximately 65 million years ago

OPEN QUESTION: Whether the post-extinction flood of dead plant material created the new ecological opportunities Senanayake suggests, raised as a possibility, not proven

Why it matters

This giant fungus shows how life can seize new chances after a catastrophe, and studying its ancient family helps scientists understand how forests and fungi recovered after the dinosaurs disappeared.

Test what you remember

1. What nickname was given to the newly discovered fungus?

- (A) Dino-cap
- (B) D-rex
- (C) King fungus
- (D) Yunnan giant

2. What is the species called?

- (A) Xylariales
- (B) Dianjunus rex
- (C) Dianjunaceae
- (D) Ascomycota

3. How tall can its upright reproductive structures reach?

- (A) 65 cm
- (B) 50 cm (20 inches)
- (C) 20 cm
- (D) 2.2 kg (4.9 pounds)

4. Where did the team find the fungus?

- (A) In a laboratory in England
- (B) In forests in Yunnan, China
- (C) On a beach in Turkey
- (D) In a desert in Africa

5. What method did scientists use to estimate when the family first appeared?

- (A) A weather record
- (B) A telescope survey
- (C) A fossil dig
- (D) A molecular clock

6. What did Dr Senanayake suggest the extinction may have created?

- (A) A colder climate in Yunnan
- (B) A new kingdom of fungi
- (C) New ecological opportunities for wood-associated decomposing fungi
- (D) A mass extinction of fungi

Answer key (try the quiz first!): 1: B 2: B 3: B 4: B 5: D 6: C

Questions to think about

1. Dr Indunil C Senanayake, Kunming Institute of Botany: After the extinction, forests changed dramatically and large amounts of dead plant material became available, which may have created new ecological opportunities for wood-associated decomposing fungi.

MY THOUGHTS (at least 20 words):